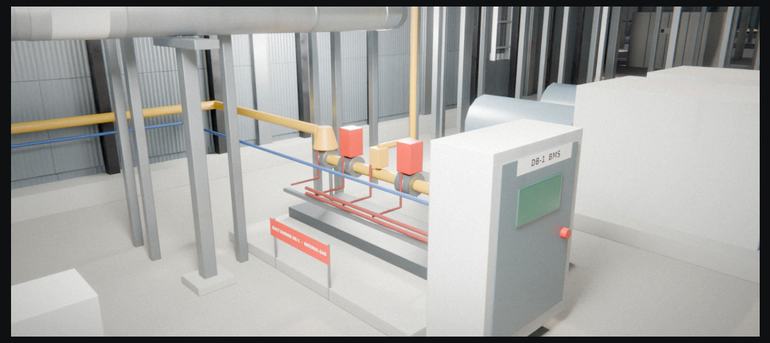
BOILERS: GREENFIELD & BROWNFIELD

One cable partner from the new HRSG to the package boiler to the outage that replaces a module: the same products, staged and pulled the same way.



01 · GREENFIELD

HEAT-RECOVERY BOILER

Behind every gas turbine: three-pressure HRSG with SCR and a duct burner. Feed pumps at 13.8 kV, drum-level and thermocouple runs, burner management in a Class I Div 2 skid.

705
CABLES PER HRSG TRAIN

156,000
FT OF CABLE PER TRAIN



02 · GREENFIELD OR RETROFIT

GAS-FIRED PACKAGE BOILER

The plant's auxiliary boiler, the same package sold to industry: burner and BMS, gas train with double block and bleed, FD fan on a VFD, feedwater pumps, its own MCC e-house.

83
CABLES PER PACKAGE BOILER

42,000
FT OF CABLE



03 · BROWNFIELD

MODULE REPLACEMENT OUTAGE

An HP evaporator module lifted out through the opened roof. The cable work rides the same outage window, so it has to be staged, pre-terminated and pulled fast.

Outage
CRITICAL PATH

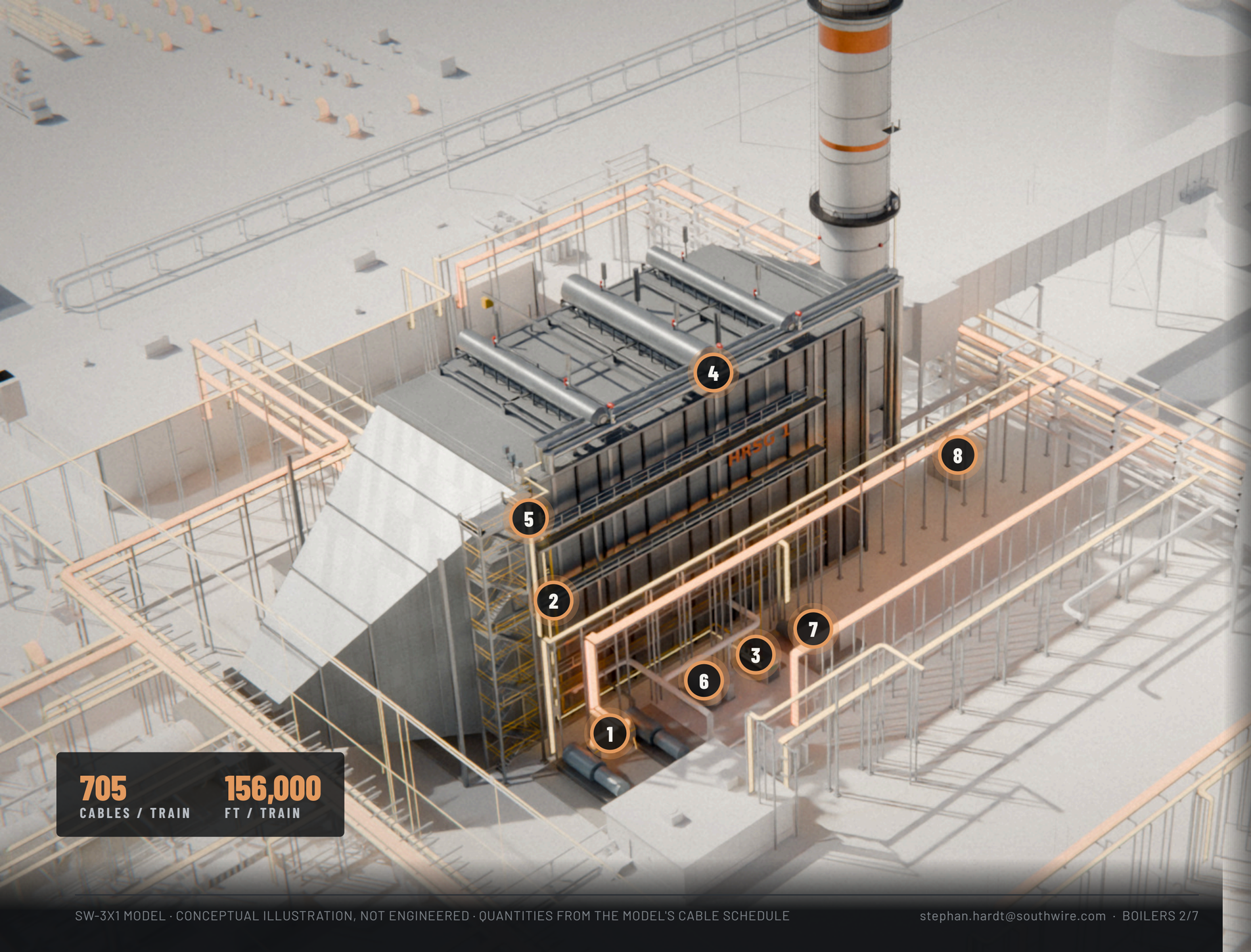
SIMpull
+ PRE-TERMINATED HARNESSES

CABLE BY BOILER TYPE

	MV-105 15 KV FEED PUMPS	TC-ER 600 V POWER & VFD	ARMOR-X MC-HL GAS TRAINS (C1D2)	CONTROL TC-ER	INSTRUMENTATION PLTC · TC EXT.	SIMPULL · REEL ROLLERS	PRE-TERMINATED HARNESSES
HRSG + DUCT BURNER	●	●	●	●	●	●	●
PACKAGE BOILER		●	●	●	●	●	●
REPLACEMENT OUTAGE	○	●	●	●	●	●	●

● in scope ○ when the outage includes feed-pump or MV work

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED · QUANTITIES FROM THE MODEL'S CABLE SCHEDULE



705 CABLES / TRAIN

156,000 FT / TRAIN

BOILER SOLUTIONS · HRSG

THE HRSG, EVERY CABLE

One HRSG train and everything wired to it: cable in copper (MV) and amber (LV, control, instruments), the rest of the plant faded.

1. **FEED PUMPS, 13.8 KV**
   MV terminal boxes off the drop tray
   MV-105 15 kV [1]
2. **DUCT BURNER**
   gas header, 8 runners, igniters and flame scanners
   Control TC-ER [5] Instrumentation PLTC [6]
3. **DUCT-BURNER SKID AND BMS**
   double block and bleed, Class I Div 2
   ARMOR-X MC-HL [3] Control TC-ER [5]
4. **DRUMS AND ROOF VALVES**
   drum-level transmitters, motor-operated stop valves
   TC-ER power [2] Instrumentation PLTC [6]
5. **RISER TRAY TO THE ROOF**
   control, instrument and thermocouple cable
   Control TC-ER [5] Instrumentation PLTC [6] Thermocouple ext. [7]
6. **CHEMICAL FEED**
   metering-pump motors
   TC-ER power [2]
7. **SCR DILUTION-AIR BLOWERS**
   480 V motors
   TC-ER power [2] MC-HL VFD [4]
8. **CEMS SHELTER**
   analyzers and stack probes
   Instrumentation PLTC [6]

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BOILER SOLUTIONS · ELEVATION

THE TRAIN IN ELEVATION

The same train at a low angle from the west: e-house, the tray into the marshalling box, the ammonia grid, drums and stack. Cable in copper and amber, the rest faded.

1. **HRSG E-HOUSE R2A**
   MCCs and DCS I/O for the unit
   TC-ER power [2] Control TC-ER [5]
2. **EL 42 CONTROL TRAY**
   into the west-wall marshalling box
   Control TC-ER [5] Instrumentation PLTC [6]
3. **AMMONIA INJECTION GRID**
   AIG header and lances, SCR
   Control TC-ER [5] Instrumentation PLTC [6]
4. **BLOWDOWN TANK**
   level and valve control
   Control TC-ER [5] Instrumentation PLTC [6]
5. **DRUMS AND ROOF VALVES**
   level transmitters, MOVs
   TC-ER power [2] Instrumentation PLTC [6]
6. **STACK AND CEMS PORTS**
   sample lines, probes
   Instrumentation PLTC [6]
7. **GT EXHAUST TO THE HRSG**
   inlet duct thermocouples
   Thermocouple ext. [7]

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INSIDE THE ROOM

Roof and two walls lifted off the HRSG 1 e-house: every cable leg, transformer to section.

1. **13.8 KV PRIMARY**
   MV-105 off the EL 36 tray into T-R2A · 3 cables, 660 ft
   MV-105 15 kV [1]
2. **480 V SECONDARY**
   10 sets of 4 x 1/C 500 kcmil · 40 cables, 4,800 ft
   XHHW-2 1/C 500 kcmil
3. **MAIN BREAKER**
   the 40 cables drop into LC-R2A
   XHHW-2 1/C 500 kcmil
4. **LC-R2A FEEDERS**
   to MCCs, VFDs, UPS and panels
   TC-ER power [2]
5. **MCC DROPS**
   power and control into each bucket
   TC-ER power [2] **Control TC-ER** [5]
6. **VFD AND MCC-2**
   SCR blowers, chemical feed, blowdown
   TC-ER power [2] **MC-HL VFD** [4]
7. **MARSHALLING**
   home-run multipairs from the field JBs
   Instrumentation PLTC [6] **Thermocouple ext.** [7]
8. **OUT TO THE FIELD**
   power via the north wall, instruments via the roof
   TC-ER power [2] **Control TC-ER** [5] **Instrumentation PLTC** [6]
9. **UPS, DC AND PANELS**
   125 V DC, 120 V AC UPS, lighting
   TC-ER power [2] **Control TC-ER** [5]



Inside: LC-R2A and MCC-R2A-1 (right), VFDs (left), trays overhead

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SWITCHGEAR ROOM TO THE HRSG

Every leg, from the 13.8 kV breaker to the field devices, by cable application: Southwire product (numbers as on the offering slide), cable count and footage from the model's schedule.

GSU-1 · UAT-1
21 / 13.8 kV



SWITCHGEAR ROOM R1

SWGR-13.8-1

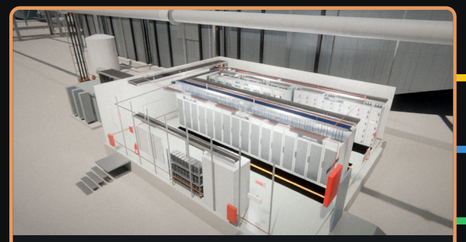
13.8 kV metal-clad: the feed-pump breakers and the T-R2A transformer feeder

MV-105 15 KV · 6 CABLES · 8,400 FT · 1

PRIMARY · MV-105 · 3 CABLES · 660 FT

T-R2A 13.8/ 0.48

SECONDARY · 1/C 500 KCMIL · 40 CABLES · 4,800 FT



HRSG E-HOUSE R2A

LC-R2A · MCCS · VFDS · DCS

480 V switchgear, starters, drives, UPS / DC, marshalling

In the room: lighting and receptacles, 2,280 ft THHN; panel feeder, control, fire-alarm loop, 330 ft

MAIN CONTROL ROOM
DCS servers, CEMS data

DCS FIBRE · 1 CABLE · 730 FT

FEED PUMPS BFP-1A / 1B, 13.8 KV
MV-105 15 kV · 6 cables · 8,400 ft 1

MOTORS, MOVS, HEATERS: SCR BLOWERS, CHEMICAL FEED, BLOWDOWN, CEMS
TC-ER 600 V power · VFD · 77 cables · 17,800 ft 2 · 4

VALVES, MOVS, BURNER MANAGEMENT, LOCAL STATIONS
Control TC-ER · ARMOR-X MC-HL · 96 cables · 21,140 ft 5 · 3

FIELD JUNCTION BOXES TO TRANSMITTERS, FLAME SCANNERS, RTDS
Instrumentation PLTC / ITC · RTD · 439 cables · 86,600 ft 6

GAS-PATH, TUBE-METAL AND DUCT THERMOCOUPLES
Thermocouple extension · 84 cables · 19,680 ft 7

T-R2A FANS, CONTROLS, TEMPERATURE / GAS MONITORING
TC-ER power · control · PLTC · 38 cables · 3,600 ft 2 · 5 · 6

CEMS SHELTER: DATA TO THE CONTROL ROOM, DETECTION, LIGHTING
Fibre · fire alarm · THHN · 3 cables · 2,580 ft —

THE HRSG 1 TRAIN: 705 CABLES, ~156,000 FT

+ FEEDING THE ROOM: 86 CABLES, 12,400 FT



R2A also serves the GT1 unit loads, so the T-R2A primary and secondary, the transformer auxiliaries, the DCS fibre and the room's own wiring are shown here but are not in the train total. Schedule rows: A-MV-0015, A-P-0422 and the rest by tag.

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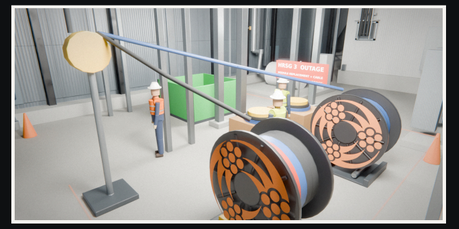
HRSG 3: an HP evaporator module lifted out through the opened roof by a crawler crane; the new module waits on a lowboy on the ring road.

BOILER SOLUTIONS · REPLACEMENT

THE OUTAGE CLOCK IS THE COST

On a boiler or HRSG replacement every hour of the outage costs. Cable rides the same window, so it moves off the critical path.

- **STAGE BEFORE THE TRIP** cut-to-length reels on SIMpull Cradles and reel rollers, labelled per pull, at the unit before the outage starts
- **PRE-TERMINATE** factory-made harnesses with QR tags land on the marshalling boxes: test, not terminate, on the critical path
- **PULL CLEAN** SIMpull reels and a sheave at the HRSG wall: one crew, no reel jacks, tension recorded
- **CUT BACK AND RECYCLE** old cable segregated into recycling bins as the module comes out
- **CONVERSIONS** new burner management and gas trains in Class I Div 2 run in ARMOR-X MC-HL



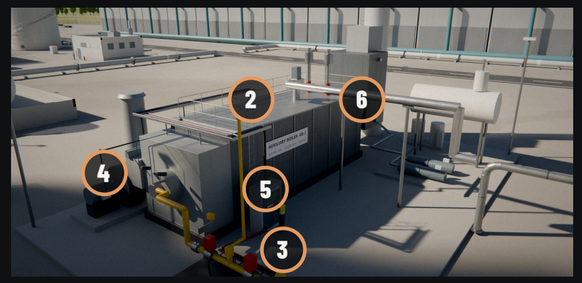
Cable work at the HRSG e-house



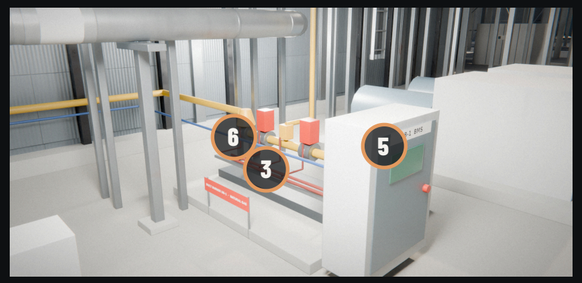
Old module on the hook

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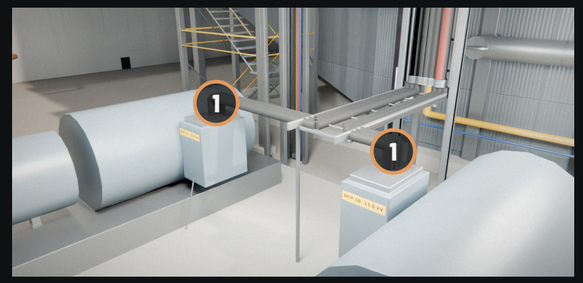
THE CABLE AT EVERY SPOT



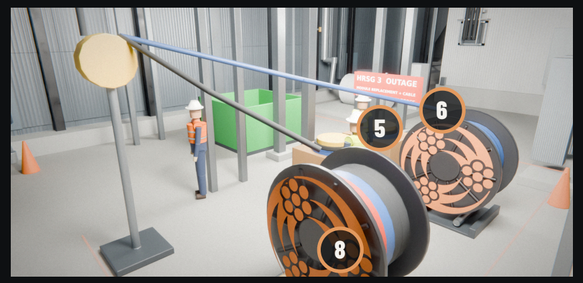
Package boiler AB-1



HRSG duct-burner skid



HRSG feed pumps, 13.8 kV



Outage cable work, HRSG 3

	PRODUCT	WHERE IT RUNS	SPEC	SOUTHWIRE STOCK # · SIZE	SPEC SHEET
1	**MV-105 15 KV** 1/C Cu, 220 mil NL-EPR 133 %, tape shield, SIMpull PVC	Feed-pump motors, 13.8 kV	46302	95604599 · 500 kcmil	southwire.com ›
2	**TYPE TC-ER POWER** 3/C Cu XHHW-2 + ground, PVC, 600 V	Fans, pumps, MCC feeders	45252	95305999 · 12 AWG 3/C	southwire.com ›
3	**ARMOR-X MC-HL** Cu XLPE XHHW-2, welded armor, PVC, 600 V, Class I Div 2	Gas trains, burner skids	45352	89052799 · 8 AWG 4/C	southwire.com ›
4	**ARMOR-X MC-HL VFD** 3/C Cu XLPE XHHW-2 + ground, VFD rated	FD fan on its drive	45352	89051899 · 3/0 3/C	southwire.com ›
5	**CONTROL TC-ER** Cu XLPE XHHW-2, 600 V	BMS panels, valves, MOVs	45053	89067701 · 14 AWG 5/C	southwire.com ›
6	**INSTRUMENTATION PLTC / ITC** 300 V, shielded pairs + overall shield (SPOS)	Flame scanners, drum level, transmitters	43303	per pair count	southwire.com ›
7	**THERMOCOUPLE EXTENSION** PVC, PLTC multi-pair (KX on request), 105 °C	HRSG gas-path and tube temperatures	42345	per construction	cabletechsupport.southwire.com ›
8	**SIMPULL CRADLE** for 51 / 61 in SIMpull Reels, one-person payoff	Outage staging and pulls	SPR-CRDL	67606001	southwire.com ›

Stock numbers are one catalogue size per product, from Southwire's product listings; every size and construction is on the same SPEC. Confirm current stock with Southwire customer service before quoting.

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